

Topics in Politics: Applied Statistical Programming

L32 5625

Lectures: Tuesdays
10:00 am – 12:00 pm
Seigle 207

Instructor Information

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Office Hours: Tuesdays, from 2:00 to 4:00 pm or by appointment

Course description

This course is designed for advanced PhD students who want to move beyond using existing software packages and learn how to build, debug, and improve statistical and computational methods for their own research. The main goal is to connect statistical principles with their computational implementation. Students will work with modern algorithms, develop strategies for debugging and optimizing code, and learn how to write programs that are clear, efficient, and reproducible. A central part of the course will be hands-on programming. Students will learn how to approach complex coding problems, improve the performance of their code, and use AI tools responsibly as part of the programming workflow. The course will emphasize not only specific techniques, but also the broader skills needed to adapt as computational tools and research practices continue to change. By the end of the course, students should be better prepared to design custom computational solutions, work with complex data, and use programming as a key component of methodological and empirical research.

Learning objectives

By the end of this course, you should be able to:

1. Implement computational methods for applied research in Political Science using AI code-generation assistants to optimize algorithmic efficiency.
2. Create custom statistical programming tools to address methodological problems faced by applied researchers by using AI debugging tools to refine code pipelines.
3. Refactor existing scripts into scalable, vectorized code, utilizing AI to identify bottlenecks and generate optimization strategies.
4. Publish reproducible statistical R packages with comprehensive documentation generated and verified using AI documentation tools.
5. Use version control via collaborative platforms (e.g., GitHub) and integrated AI tools to manage workflows and code integration.

Course preparation

This is not an introductory R course. Students should enter the course with working proficiency in R, including basic data manipulation, visualization, relational data, and string operations. Tools such as `ggplot2`, `data.table`, and regular expressions will be used as needed, but they will not be taught as stand-alone units. Short review materials will be provided when useful. The emphasis of the course is on implementing, debugging, evaluating, and scaling statistical algorithms rather than on learning the basic syntax of R.

Assistant Instructor

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Course Requirements

Your final grade is based on lab attendance, the problem sets, and a final exam:

- **Labs** (10 points): There will be a weekly lab session where you will work through a guided practical exercise designed to help with the understanding of the materials covered that week and the problem sets. You are required to attend ALL lab sessions. Attendance: 5 points; Lab exercises: 5 points. The grading scheme for each lab assignment is ✓ – (50% of the points), ✓ (75% of the points), ✓ + (100% of the points).

Labs: Fridays

Time: 9:00 to 9:50 am

Place: TBD

- **Problem sets** (60 points): There will be five to six problem sets throughout the semester. Each problem set will contribute equally to the final grade and will contain analytical and data analysis questions.

For this class, problem sets provide opportunities for individual learning and group collaboration.

- **Final Exam** (30 points): Covering all of the course materials.

Grading Scale

A	B	C	D	F
	B+: ≥ 87	C+: ≥ 77	D+: ≥ 67	F : < 60
A : ≥ 94	B : ≥ 83	C : ≥ 73	D : ≥ 63	
A–: ≥ 90	B–: ≥ 80	C–: ≥ 70	D–: ≥ 60	

Textbooks

There is no single textbook for this course. However, you may find the relevant parts of the following textbooks useful. You do NOT need to buy any of them.

Wickham, Hadley. 2019. *Advanced R*. (2nd Edition) CRC Press. <https://adv-r.hadley.nz/>

Wickham, Hadley. 2015. *R Packages: Organize, Test, Document, and Share Your Code*. O'Reilly.

Eddelbuettel, Dirk. 2013. *Seamless R and C++ Integration with Rcpp*. Springer. (Check SpringerLink through the library.)

Optional review materials

Students who would benefit from a refresher on data manipulation, visualization, relational data, or other foundational R skills may consult:

Wickham, Hadley and Garrett Golemund. 2017. *R for Data Science*. O'Reilly. <https://r4ds.had.co.nz/>

AI Policy for This Class

This course treats AI tools as part of the modern programming environment. You are allowed, and in some cases encouraged, to use AI assistants to help you learn, debug, document, and improve your code. However, AI should support your learning, not replace it. The goal of this course is for you to become better at programming, not merely to submit code that runs.

On the AI Assessment Scale, this course operates at an “AI-Assisted & Monitored” level. This means AI should act as a personalized tutor or code reviewer, rather than a co-author.

You are encouraged to use AI tools for:

- Asking for explanations of programming concepts, help understanding error messages, or unfamiliar functions.
- Getting help debugging code that you have written.
- Asking for suggestions about how to improve code efficiency, readability, or organization.
- Generating isolated examples (not part of your assignment) to help you understand a concept.
- Using AI to help you outline comments or documentation of your code.
- Comparing alternative approaches to a programming problem.

However, you may not use AI tools in ways that substitute for your own critical thinking and implementation. In particular:

- Do not copy and paste raw code blocks or complete functions generated by an AI assistant directly into your submission.
- Do not submit code or documentation you do not understand.
- Do not ask AI to complete an entire homework assignment for you.
- Do not use AI to bypass the learning process.
- Do not present AI-generated explanations, interpretations, or analyses as if they were entirely your own.

For assignments in this course, you should be able to explain every line of code you submit. Note that I may ask you to walk through your code, describe why it works, explain your debugging process, or justify your implementation choices. If you cannot explain what your code does, then you have not completed the assignment in the spirit of the course, even if the code produces the correct output.

AI Use and Attribution. In alignment with Washington University’s standards of academic integrity, you bear full responsibility for the accuracy and originality of your submissions, regardless of whether AI tools were used.

When you use AI tools, you must disclose that use. This does not mean every quick syntax query must be reported. However, if AI helped you restructure a function, diagnose a complex bug, optimize your logic, or draft a meaningful part of your code’s documentation, you must include a short note with your submission. Please include this disclosure as a commented block at the very top of your main R script file.

Examples of proper attribution:

```
# DISCLOSURE: I used [AI tool name] to help understand an error message regarding
a numerical optimization solver. I then revised the code myself.
```

or:

```
# DISCLOSURE: I used an AI assistant to compare two approaches to parallelizing a
bootstrap procedure. The final implementation, testing, and adjustments are entirely
my own.
```

In some assignments, you may be asked to submit a brief AI-use log that includes the prompts you used, the responses you received, and a short reflection on how the interaction affected your work.

Other Class Policies

- **Attendance:** is a course requirement and critical to having a lively conversation, and if possible, try to avoid missing any classes.
- **Software and hardware:** We will use the R statistical programming language and RStudio throughout the course. We will also use Git and GitHub for version control and collaboration. Some advanced tasks may be taught assuming a Mac environment. Students using Windows should work with the instructor or TA early in the semester to make sure their computing environment is properly configured.

Once you are set up with R, RStudio, Git, and the required packages, do not update your operating system during the semester unless necessary. Operating system updates can break package installations, compilers, Git configurations, or other tools we rely on in class.

- **Technology in the classroom:** You will frequently use computers in this course. Please use them only for class-related purposes during class. Programming requires sustained attention, and frequent multitasking will make it harder for you and your peers to learn. Please avoid checking email, social media, messaging apps, or unrelated websites during class.

Phones should be put away before class begins.

- **Academic Integrity:** Academic integrity is vital to our shared learning community. To maintain a fair environment and protect the value of your education, the university's academic integrity standards apply to all work in this course, including work completed with AI assistance.

Specific rules for this course:

- You may discuss homework and programming problems with classmates unless otherwise instructed.
- You may work in small groups when collaboration is explicitly allowed.
- Unless otherwise stated, each student must prepare and submit their own work.
- Homework and in-class assignments are generally open-book and open-notes.
- Restricted assessments may only involve the resources explicitly permitted by the instructor.
- Please reach out to the instructor or TA if you have questions about whether a particular use of or collaboration with AI is allowed. We are here to help you navigate these tools safely.

The guiding principle is simple: AI can help you learn, but it cannot do the learning for you. Your submitted work should reflect your own understanding, judgment, and effort.

- **Religious observances:** Some students may wish to participate in religious observances during this semester. If you have a religious observance that conflicts with your participation in the course, please meet with the instructor before the end of the second week of the semester to discuss appropriate accommodations.

- **COVID-19 Health and Safety Protocols**

Students experiencing symptoms consistent with COVID-19 or concerned about possible exposure should contact Habif Health and Wellness Center (314 935-6666) to arrange for testing as indicated. If instructed by Habif to quarantine or isolate, students should notify their instructor as soon as possible by forwarding the email they received from Habif. Any accommodation needs for COVID-related absences not covered in an instructor's standard course policies should be discussed between the student and instructor.

While on campus, it is imperative that students follow all public health guidelines established to reduce the risk of COVID-19 transmission within our community. The full set of University protocols can be found at <https://covid19.wustl.edu/health-safety/>. This includes:

Masking remains a valuable tool in the mitigation of COVID-19, particularly in light of new and emerging variants. Students and instructors are encouraged to treat requests to mask with care and consideration, keeping in mind that some individuals may be at a higher risk, caring for others at a higher risk, or feeling less comfortable in a mask-optional environment. Based on the monitoring of regional and campus conditions, a mask requirement may be implemented as needed. For current masking policies, see the WashU Together website.

Students with disabilities for whom masked instructors or classmates create a communication barrier are encouraged to contact Disability Resources (www.disability.wustl.edu) or talk to their instructor for assistance in determining reasonable adjustments. Adjustments may involve amplification devices, captioning, or clear masks but will not allow for the disregard of mask policies should a requirement be in place.

While distancing requirements have been removed for vaccinated students, those who are not fully vaccinated are strongly encouraged, for their own health, to maintain a distance of 6 ft from others in the classroom. If you are not able to be vaccinated or have conditions that may put you at increased risk of failed immunity and classroom activities would bring you in frequent proximity to other students, contact your instructor to discuss alternatives.

Practicing healthy personal hygiene, including frequent handwashing with soap and warm water for at least 20 seconds and/or using hand sanitizer with at least 60% alcohol.

- **Reporting Sexual Assault and Harassment**

If a student discusses or discloses an instance of sexual assault, sex discrimination, sexual harassment, dating violence, domestic violence or stalking, or if a faculty member otherwise observes or becomes aware of such an allegation, the faculty member will keep the information as private as possible, but as a faculty member of Washington University, they are required to immediately report it to the Department Chair or Dean or directly to Ms. Cynthia Copeland, the University's Associate Title IX Coordinator, at (314) 935-3411, cm-copeland@wustl.edu. Additionally, you can report incidents or complaints to the Office of Student Conduct and Community Standards or by contacting WUPD at (314) 935-5555 or your local law enforcement agency. See: Title IX

- **Disability Resources (DR)**

WashU supports the right of all enrolled students to an equitable educational opportunity, and strives to create an inclusive learning environment. In the event the physical or online environment results in barriers to the inclusion of a student due to a disability, they should notify the instructor as soon as possible.

Disabled students requiring adjustments to equitably complete expectations in this course should contact WashU's Disability Resources (DR), and engage in a process for determining and communicating reasonable accommodations. Because accommodations are not applied retroactively, DR recommends initiating requests prior to, or at the beginning of, the academic term to avoid delays in accessing accommodations once classes begin. Once established, responsibility for disability-related accommodations and access is shared by Disability Resources, faculty, and the student.

Disability Resources: <http://www.disability.wustl.edu/>; 314-935-5970

- **Statement on Military Service Leave**

Washington University recognizes that students serving in the U.S. Armed Forces and their family members may encounter situations where military service forces them to withdraw from a course of study, sometimes with little notice. Students may contact the Office of Military and Veteran Services at (314) 935-2609 or veterans@wustl.edu and their academic dean for guidance and assistance. See: <https://veterans.wustl.edu/policies/policy-for-military-students/>.

- **Preferred Name and Personal Pronouns**

Washington University in St. Louis recognizes that many students prefer to use names other than their legal ones to identify themselves. In addition, in order to affirm each person's gender identity and lived experiences, it is important that we ask and check in with others about pronouns. This simple effort can make a profound difference in a person's experience of safety, respect, and support. See: <https://students.wustl.edu/pronouns-information/>, <https://registrar.wustl.edu/student-records/ssn-name-changes/preferred-name/>.

- **Emergency Preparedness**

Before an emergency, familiarize yourself with the building(s) that you frequent. Know the layout, including exit locations, stairwells and the Emergency Assembly Point (EAP). Review the "Quick Guide for Emergencies" that is found near the door in many classrooms and main lobby areas of buildings for specific emergency information and instructions. For additional Information and EAP maps, visit <https://emergency.wustl.edu/>. To ensure that you receive emergency notifications, make sure your information and cell phone number is updated in SIS, and/or download the WashU Safe app and enable notifications.

To report an emergency:

Danforth Campus: (314) 935-5555

School of Medicine Campus: (314) 362-4357

North/West/South and Off Campus: 911 then (314) 935-5555

- **Academic Integrity** Effective learning, teaching, and research all depend upon the ability of members of the academic community to trust one another and to trust the integrity of work that is submitted for academic credit or conducted in the wider arena of scholarly research. Such an atmosphere of mutual trust fosters the free exchange of ideas and enables all members of the community to achieve their highest potential.

In all academic work, the ideas and contributions of others must be appropriately acknowledged, and work that is presented as original must be, in fact, original. Faculty, students, and administrative staff all share the responsibility of ensuring the honesty and fairness of the intellectual environment at Washington University in St. Louis.

For additional details on the university-wide Undergraduate Academic Integrity policy, please see: <https://wustl.edu/about/compliance-policies/academic-policies/undergraduate-student-academic-integrity-policy/>

Plan of the course

The course is organized around *statistical methods and computational problems*, rather than around isolated programming features. This is very intentional. The goal is that programming tools are

introduced when they solve a concrete methodological or computational problem, as it is often the case with your own research.

AI-assisted code development and validation: students may use AI tools to explain unfamiliar code, diagnose bugs, identify bottlenecks, suggest alternative implementations, or help draft documentation, subject to the AI policy above. The central expectation is that students evaluate those suggestions rather than accept them uncritically: AI-assisted code should be tested for correctness, benchmarked when efficiency is at issue, and compared against transparent baselines whenever possible.

Very tentative 14-week schedule

This is an ambitious course that will require substantial intellectual engagement from all of us. It will also require *flexibility*: the exact pace may evolve in response to the needs of the class, the projects, and issues raised by students. Topic-specific readings and code examples will be posted on Canvas. The schedule below should therefore be viewed as a roadmap rather than a fixed day-by-day calendar.

Week	Topic	Computational focus	Assignments
Section 1: Computational foundations			
1	Computational workflow	R/RStudio workflow; Git and GitHub; functions; reproducibility; package skeletons; responsible AI-assisted programming	
2	Efficient R	Vectorization; profiling; debugging; benchmarking; unit testing; code refactoring; AI as code reviewer	
3	Resampling: bootstrap and permutation tests	Reusable simulation functions; random-number control; transparent baseline implementations	PS 1: Resampling at scale
4	Parallel computing and scalable resampling	Parallel mapping; reproducible parallel computation; speedup and overhead; benchmark design	
Section 2: Statistical algorithms and optimization			
5	Clustering: K-means and Gaussian mixture models	Matrix operations; initialization; objective functions; optimization; convergence diagnostics	PS 2: Latent clustering algorithms
6	EM algorithm: mixture of multinomials for text	Iterative algorithms; numerical stability; stopping rules; debugging convergence; tests	
7	Pairwise comparisons: cosine similarity, LSH, and approximate search	Computational complexity; sparse representations; blocking/indexing; profiling pairwise operations	PS 3: Scaling pairwise comparisons
8	Rcpp and optimized pairwise algorithms	Moving computational kernels to C++; interfaces between R and C++; validating and benchmarking optimized code	

Week	Topic	Computational focus	Assignments
Section 3: Simulation, approximation, and modern methods			
9	Simulation for text-analysis methods	Monte Carlo design; train/test splits; out-of-sample evaluation; functional/parallel workflows; reproducible comparisons	
10	Ideal-point estimation: Bayesian approach	Posterior computation; samplers; efficient matrix code; diagnostics; numerical validation	PS 4: Approximate inference; project development begins
11	Ideal-point estimation: variational inference	Optimization-based approximation; convergence; speed/accuracy tradeoffs; comparison to Bayesian computation	
12	Embeddings	High-dimensional representations; similarity and nearest neighbors; dimensionality; scalability; evaluation	PS 5: Embeddings and active learning
13	Active learning	Uncertainty/information-based sampling; iterative data acquisition; human-in-the-loop workflows; performance evaluation	
Section 4: Research software integration			
14	Integration, package development, and project presentations	Documentation; tests; benchmarks; reproducibility; code review; AI-use reflection; software handoff	Final exam